$$\hat{H} = \hat{H}_{\mathcal{S}} + \hat{H}_{\mathcal{E}} + \hat{H}_{\text{int}} = \hat{H}_{\mathcal{S}} - \sum_i \frac{\Delta_i}{2} \hat{\sigma}_x^{(i)} + \hat{X} \otimes \sum_i g_i \hat{\sigma}_z^{(i)}, \tag{5.196}$$